Lecture 11: Early CAPM Tests and First Cracks

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Intro

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- Logic: CAPM benchmark first, data second, anomalies third, joint tests last

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Flight Plan:

- Part I: why CAPM looked compelling in theory (Sharpe 1964; Lintner 1965; Mossin 1966)
- Part II: early empirical protocols from Black, Jensen, and Scholes (1972) and Fama and MacBeth (1973)
- Part III: the first anomaly evidence (Basu 1977; Banz 1981; De Bondt and Thaler 1985)

Part I: CAPM Recall

Early Days

- In 1952: Markowitz brings us to light with “Modern Portfolio Theory’ ’
- Key insight: you should diversify your investments, correlation matters
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In the 1960s:

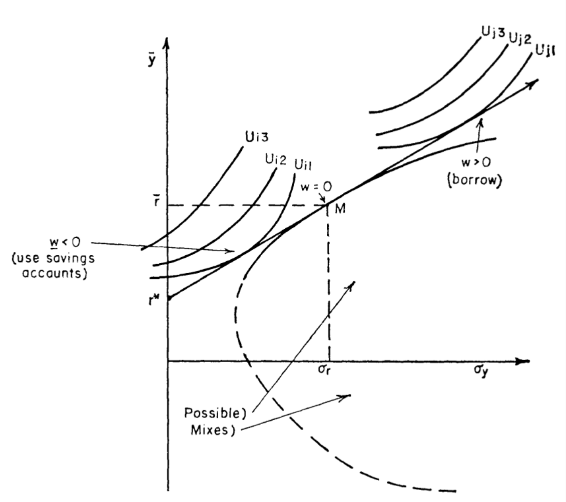
- Sharpe (1964), Lintner (1965), and Mossin (1966) build on Markowitz to get the CAPM
- Notice this *way* before the Rational Expectation revolution and the SDF framework
- These guys were The Beatles of Finance!

What CAPM Delivered

- It linked portfolio choice to equilibrium expected returns
- It delivered one sufficient statistic: market beta
- Investors are mean-variance optimizers over one period: you like returns, dislike variance
- Homogeneous beliefs and frictionless trading are assumed
- Everyone can borrow/lend at one risk-free rate and hold the same assets
- Important: different investors will have different preferences!

Mean-Variance Frontier and Tangency Portfolio

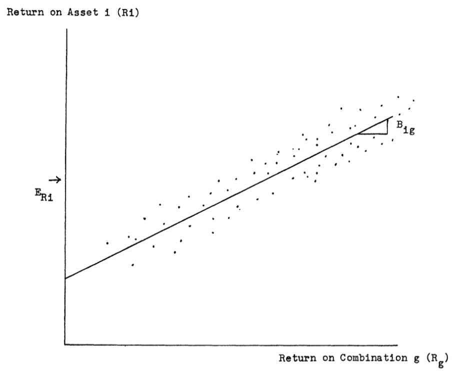
- Efficient portfolios are combinations of the risk-free asset and *one* tangency portfolio
- Tangency portfolio: highest Sharpe-ratio risky portfolio
- Investors differ only in how much risk they scale, not in risky composition
- They will *not* have the same overall portfolio!



Security Market Line

- Expected excess returns are linear in market beta
- Only systematic market covariance is priced
- Idiosyncratic risk earns no premium

$$\mathbb{E}[R_i - R_f] = \beta_i \, \mathbb{E}[R_m - R_f], \quad \beta_i \equiv \frac{\text{cov}(R_i, R_m)}{\text{var}(R_m)}$$



CAPM as SDF/Beta Representation (Bridge to Lecture 4)

$$m_{t+1} = a - bR_{m,t+1}$$

- CAPM is one special case of the general SDF framework from Lecture 4
- If the SDF is affine in market returns, beta pricing follows mechanically
- This gives a clean bridge between no-arbitrage language and equilibrium CAPM
- The idea of Bansal and Yaron (2004) to think about a "wealth portfolio" is much older than you think!

Testable Implications for Data

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \varepsilon_{i,t}$$

- Cross-section: higher beta assets should have higher average returns
- The relation should be approximately linear
- Exact CAPM implies $\alpha_i = 0$ for all test assets
- Nonzero alpha patterns indicate systematic model failure
- Important: nothing else should be added to the right-hand side!

Questions?

Black (1972): Zero-Beta CAPM

$$\mathbb{E}[R_i] = \mathbb{E}[R_z] + \beta_i(\mathbb{E}[R_m] - \mathbb{E}[R_z])$$

- Black (1972) relaxes unrestricted risk-free borrowing and lending
- A zero-beta portfolio replaces R_f as the intercept object
- The linear beta-pricing equation survives with a different baseline return

Zero-Beta CAPM: Derivation Intuition

$$\mathrm{cov}(R_z, R_m) = 0, \qquad \beta_i \equiv \frac{\mathrm{cov}(R_i, R_m)}{\mathrm{var}(R_m)}$$

- Without risk-free borrowing/lending, the efficient set is generated by risky portfolios only
- The relevant baseline return is the expected return on the portfolio orthogonal to R_m
- Orthogonality gives the same slope object (β_i), but a different intercept
- Intercept shifts from R_f to $\mathbb{E}[R_z]$

Zero-Beta Assets Are Cool Again

- The interest rates we observe on the data are a mix of many things
- Policy decisions + expectations about the future + risk premia + market frictions
- In the CAPM world, all we need is a zero-beta asset to get the main implications...

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More recently:

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Even more recently:

- Kocherlakota (2025) analyzes the theoretical price of sovereign debt discount by the zero-beta rate

Questions?

Part II: Early CAPM Tests

From Theory to Data: What Early Tests Actually Did

- Black, Jensen, and Scholes (1972) decided to “draw” the Security Market Line
- Do high-beta stocks earn higher average returns?
- Is this increase proportional to the market risk premium?
- By the way, “return on wealth” became “broad equity index return” really fast (see Roll (1977))

From Theory to Data: What Early Tests Actually Did

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- First empirical issue: how to realiably measure β_i? Returns are noisy!
- Solution: group stocks into portfolios sorted on *past betas* to reduce noise

Black-Jensen-Scholes (1972): Design + Regression Setup

- In a given time t, for N stocks, estimate $[\beta_1, ..., \beta_N]$:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \varepsilon_{i,t}$$

- Create 10 portfolios sorted on individual past betas
- Hold each portfolio for one year and redo sorting process every year
- At the end, you have 10 time series of portfolio returns

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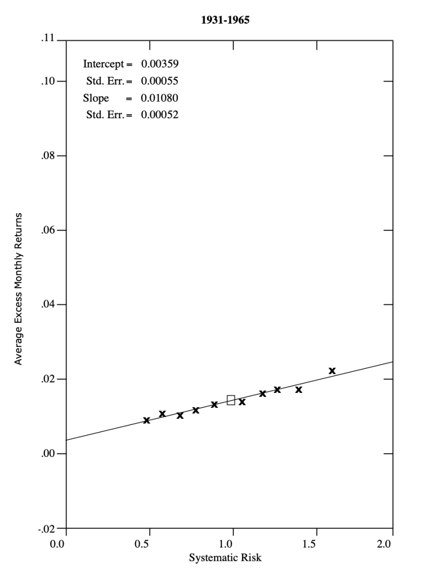
- Create 10 portfolios sorted on individual past betas
- Hold each portfolio for one year and redo sorting process every year
- At the end, you have 10 time series of portfolio returns
- Redo the same regression for each of the 10 portfolios to get 10 alphas and 10 betas
- Finally: estimate γ_0 and γ_1 from:

$$\mathbb{E}[R_{p,t} - R_{f,t}] = \gamma_0 + \gamma_1 \beta_p + u_{p,t}$$

- Important: this is a **cross-sectional** regression!

BJS Main Findings

- Beta and average returns are positively related
- Low-beta portfolios earn "too much"
- High-beta portfolios earn "too little"
- $\hat{\gamma}_0 > 0$, $\hat{\gamma}_1$ is small



BJS Regression Evidence

	Time Period				
	Total Period	*Subperiods*			
	1/31–12/65	1/31–9/39	10/39–6/48	7/48–3/57	4/57–12/65
$\hat{\gamma}_0$	0.00359	-0.00801	0.00439	0.00777	0.01020
$\hat{\gamma}_1$	0.0108	0.0304	0.0107	0.0033	-0.0012
$\gamma_1 = \overline{R}_M$	0.0142	0.0220	0.0149	0.0112	0.0088
$t(\hat{\gamma}_0)$	6.52	-4.45	3.20	7.40	18.89
$t(\gamma_1 - \hat{\gamma}_1)$	6.53	-4.91	3.23	7.98	19.61

- This was taken as a partial victory: higher beta, higher returns
- The relationship is flatter than expected...
- This is in fact support for the zero-beta CAPM!

Quick Look Into the Future

- Frazzini and Pedersen (2014) showed this “anomaly” is still alive, and true across international markets
- Also true in bond markets and commodities!
- They provide a model and a theory to explain it: embedded leverage

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- Frazzini and Pedersen (2014) showed this “anomaly” is still alive, and true across international markets
- Also true in bond markets and commodities!
- They provide a model and a theory to explain it: embedded leverage
- Some investors cannot lever up... high-beta stocks provide leverage
- They bid up the prices, so returns tend to be lower than CAPM would predict
- Fun fact: they show how Warren Buffet has been buying low-beta stocks for decades!

Questions?

Fama-MacBeth (1973): Why Two Passes?

- Exposure estimation and risk-price estimation are different statistical problems
- Two-pass procedure separates those two steps cleanly (Fama and MacBeth 1973)
- It produces a time series of risk-price estimates for inference
- This became the default empirical workflow in cross-sectional asset pricing

Fama-MacBeth Pass 1 (Estimate Betas)

$$R_{i,t} - R_{f,t} = a_i + b_i(R_{m,t} - R_{f,t}) + e_{i,t}$$

- First-pass time-series regressions estimate each asset's market beta
- Betas are treated as characteristics for the second-pass cross section
- Portfolio grouping can be used too (in fact used in Fama and MacBeth (1973))

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Pass 2:

$$R_{i,t} = \gamma_{0,t} + \gamma_{1,t}\hat{\beta}_i + u_{i,t}$$

- At each date, run a cross-sectional regression of returns on estimated betas
- This will deliver two time series of estimates: $\{\hat{\gamma}_{0,t}\}$ and $\{\hat{\gamma}_{1,t}\}$
- Average $\gamma_{1,t}$ over time to estimate the market price of beta risk
- Average $\gamma_{0,t}$ to assess the intercept restriction

Fama-MacBeth Main Findings + Limits

- A positive average beta-return relation is present
- Intercept evidence is not cleanly aligned with strict Sharpe-Lintner CAPM
- Still the backbone workflow, but with generated-regressor and beta-estimation caveats
- Conclusions depend on test assets, sample period, and market proxy quality
- Click here for huge tables
- The Roll (1977) critique is always hunting us, by the way ...

If beta is enough, why did anomalies show up so quickly?

Part III: The First Cracks

Basu (1977): Characteristics vs Beta

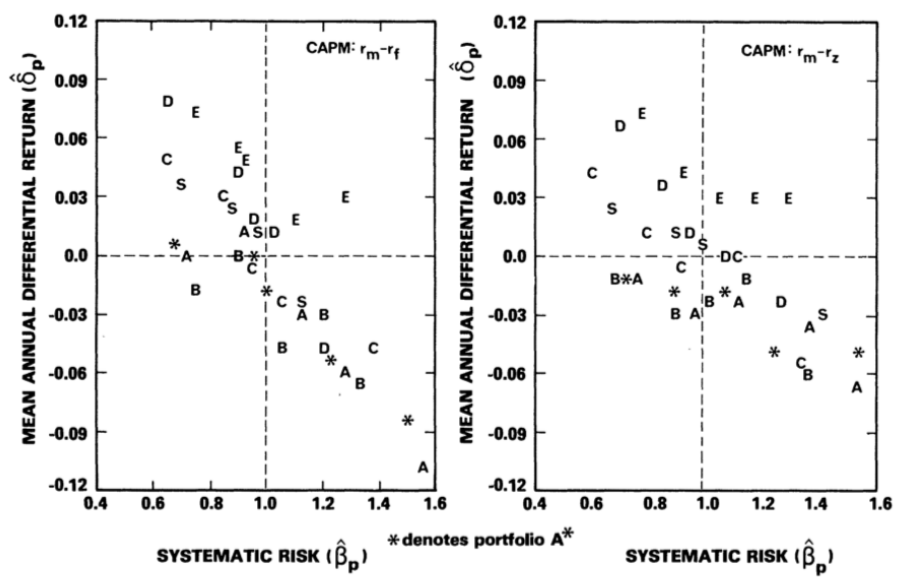
- Basu (1977) sorts stocks by earnings-to-price ratios
- High E/P portfolios earn higher average returns than low E/P portfolios
- This spread remains difficult to map into a pure one-beta CAPM story
- Early signal that characteristics may contain pricing information
- CAPM says beta should span expected-return differences; Basu says E/P also forecasts returns

Basu (1977): Main Table (A = Highest P/E, E = Lowest P/E)

PERFORMANCE MEASURES, NET OF TAX, AND RELATED SUMMARY STATISTICS
(CAPM: $r_m - r_f$; April 1957–March 1971)

	Performance Measure/Statistic[1,2]								
Portfolio	$\bar{r}_p$	$\hat{\beta}_p$	$\hat{\delta}_p$	$t(\hat{\delta}_p)$	$\bar{r}'_p/\hat{\beta}_p$	$\bar{r}'_p/\sigma(\tilde{r}'_p)$	$\rho(\tilde{r}'_p,\tilde{r}'_m)$	$\rho(\tilde{e}_t,\tilde{e}_{t+1})$	Chow Test:[3] F-Statistic
A	0.0699	1.1161	−0.0198	−2.10	0.0460	0.1094	0.9667	0.0404	2.4093
*A**	0.0703	1.0611	−0.0158	−1.61	0.0488	0.1153	0.9603	0.0827	2.1886
B	0.0647	1.0428	−0.0203	−2.86	0.0443	0.1064	0.9779	0.0360	0.4766
C	0.0810	0.9711	0.0006	0.09	0.0644	0.1543	0.9753	−0.1271	1.1642
D	0.0941	0.9451	0.0153	2.43	0.0800	0.1924	0.9788	0.0104	1.1574
E	0.1145	0.9913	0.0328	3.73	0.0969	0.2293	0.9632	0.1541	0.3168
S	0.0852	1.0126	0.0022	0.63	0.0659	0.1611	0.9941	0.1172	0.0289
F	0.0822	1.0000	—	—	0.0637	0.1566	—	—	—

Basu (1977): Main Figure



Banz (1981): Size Effect

- Banz (1981) documents higher average returns for small-cap stocks, controlling for β
- The size-return spread is hard to eliminate with market beta alone
- Small-minus-big logic later becomes a canonical factor construction. Maybe a pioneer?
- Another direct empirical crack in single-factor CAPM

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Regress excess returns of portfolios on $R_m - R_f$ and a constant (reported):

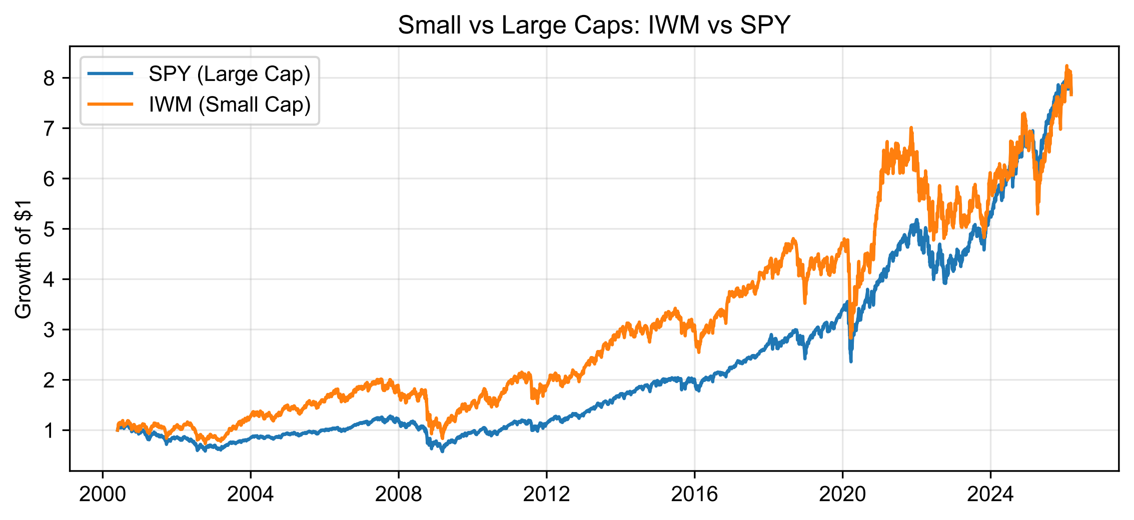
	$\tilde{\alpha}_1$ (Small Minus Large)			$\tilde{\alpha}_2$ (Small Minus Median)			$\tilde{\alpha}_3$ (Median Minus Large)		
	$n = 10$	$n = 20$	$n = 50$	$n = 10$	$n = 20$	$n = 50$	$n = 10$	$n = 20$	$n = 50$
1931–1975	0.0152 (2.99)	0.0148 (3.53)	0.0101 (3.07)	0.0130 (2.90)	0.0124 (3.56)	0.0089 (3.64)	0.0021 (1.06)	0.0024 (1.41)	0.0012 (0.85)

- The size effect was *huge*: 1.52% per month!

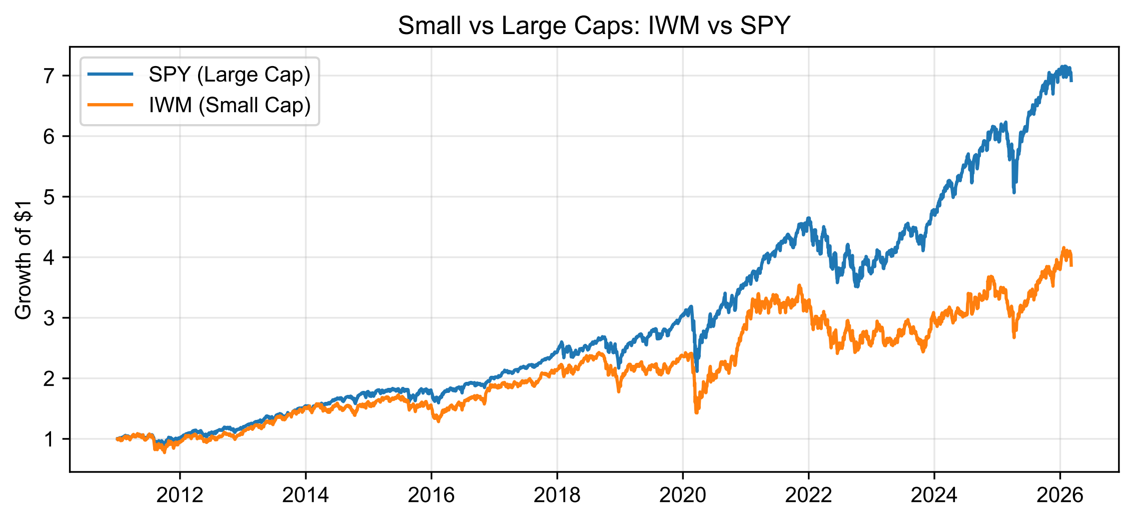
Theories Why Size Matters

- Banz (1981) proposed no theoretical explanantion, just an empirical fact
- A looming question: is it really size or is size correlated with some other fundamental factor?
- One possibility: small-cap stocks might look more like "around the money" call options
- Another possibility: liquidity (or lack thereof!)
- There are many others! Take a look on Asness et al. (2018)!

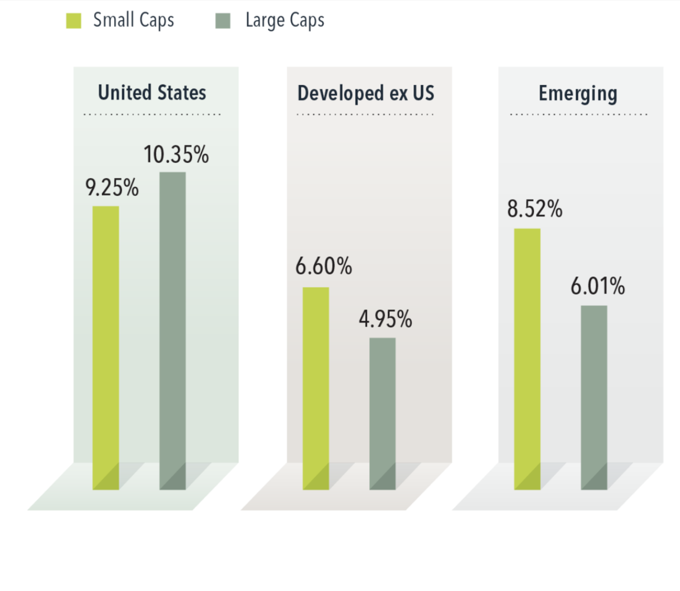
IWM vs SPY (2000-2026)



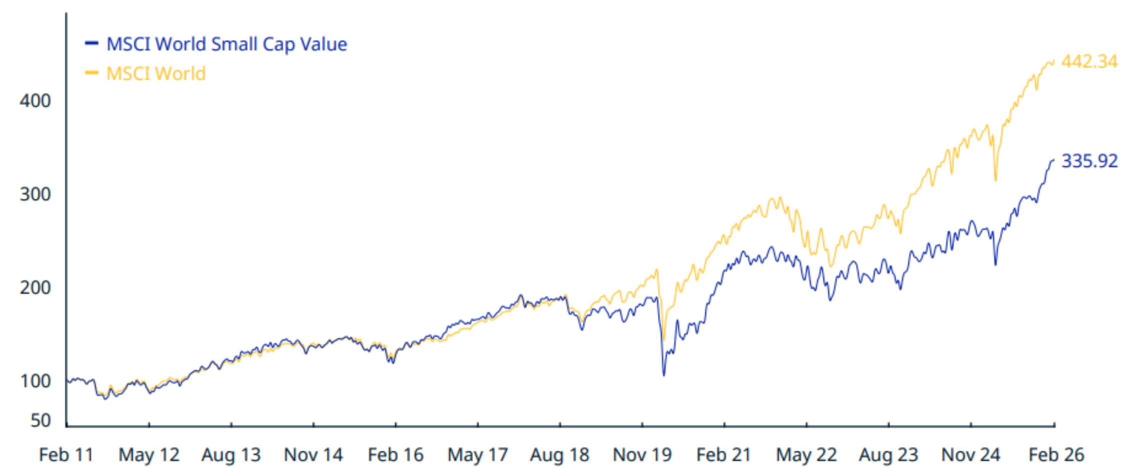
It really depends on how you measure it... (2010-2026)



From Dimensional Advisors (2005-2026)



From MSCI

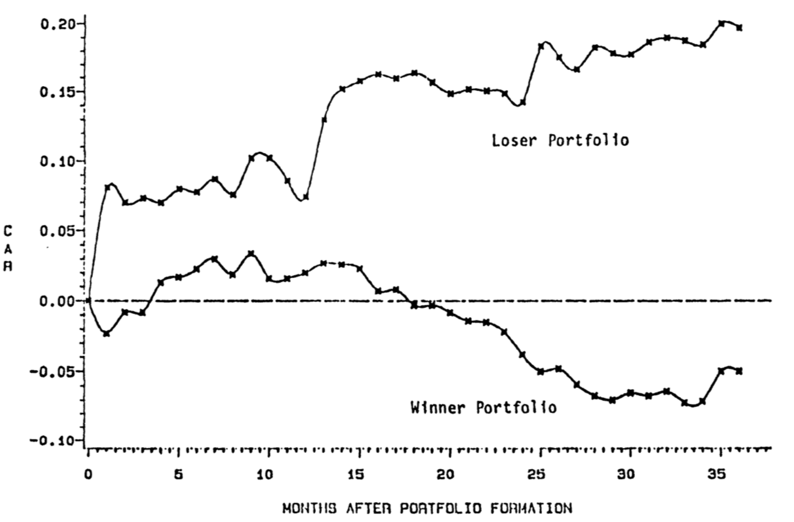


MSCI World Small Cap Value
MSCI World
442.34
335.92
400
300
200
100
50
Feb 11
May 12
Aug 13
Nov 14
Feb 16
May 17
Aug 18
Nov 19
Feb 21
May 22
Aug 23
Nov 24
Feb 26

DeBondt-Thaler (1985): Long-Run Reversal

- De Bondt and Thaler (1985) showed that long-horizon losers tend to outperform winners
- They built portfolio of stocks sorted on past performance and tracked returns
- Multi-year return reversals are difficult for static CAPM to rationalize
- Evidence of investor overreaction, similar to what the psychology literature was documenting at the time

DeBondt-Thaler (1985): Main Figure (1933-1980)



Questions?

Taking Stock

- CAPM gave a disciplined starting benchmark
- BJS and Fama-MacBeth established the empirical playbook and supported CAPM
- Basu, Banz, and DeBondt-Thaler revealed systematic cracks
- Next step: the Fama-French revolution and the multi-factor era

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